`sintaxismark`: complete user documentation

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Experimental monolithic version 0.7.2 — 2026-06-29

Contents

1 Purpose and scope

`sintaxismark` is a LaTeX package for marking syntactic constituents, grammatical functions, heads, and annotated spans directly over running text. It is designed for teaching materials, exercises, handouts, slides, and compact linguistic analyses where a full syntactic tree would be too heavy.

The experimental monolithic version consists of a single file, `sintaxismark.sty`. This file contains the classic TikZ-overlay implementation, manual multiline helpers, experimental automatic splitting commands, and the experimental LuaLaTeX grid renderer.

2 Compilation and installation

Copy `sintaxismark.sty` to the same directory as your document, or install it in a local TeX tree.

```
\documentclass{article}
\usepackage{sintaxismark}
\begin{document}
Text.
\end{document}
```

If you use `gb4e`, load `gb4e` first and disable automatic math mode:

```
\documentclass{article}
\usepackage{gb4e}
\noautomath
\usepackage{sintaxismark}
\begin{document}
\begin{exe}
\ex \SES{Juan} \PVS{\NV{compró} \OD{manzanas}}.
\end{exe}
\end{document}
```

Classic mode uses TikZ with `remember picture` and `overlay`. Compile twice if the first run places labels incorrectly. Grid mode and the automatic splitting commands require LuaLaTeX.

3 Modes of operation

1. **Classic mode**: commands such as `\spanline`, `\openbox`, `\tagbelow`, `\synul`, and the shortcut commands draw annotations directly around the text. This is the most flexible mode for short examples and fine manual control.
2. **Manual multiline mode**: helpers such as `\spanfirst`, `\spanmid`, `\spanlast`, `\boxfirst`, `\boxmid`, and `\boxlast` split a long annotation into visually separate segments.
3. **Automatic split mode**: `\autospanline` and `\autoopenbox` split plain text automatically. This is experimental and only parses plain text.
4. **Grid mode**: the `sintaxisgrid` environment parses nested sintaxismark markup, computes line breaks, and renders the whole example as one grid. It is intended to avoid diagonal lines in long multiline analyses.

4 Visual settings inherited from the classic implementation

The original spacing of `\spanline` and `\openbox` is controlled by the `/sintaxis` key family. The following defaults are the reference values used by the classic implementation.

Key	Default	Meaning
`color`	black	Annotation color.
`linew`	0.6pt	Stroke width for spans, boxes, and underlines.
`liney`	1ex	Vertical length of the legs of upper spans.

`linegap`	5.5pt	Gap between the text and the lower end of an upper span leg. The upper span line is placed at `linegap + liney`.
`labsep`	1ex	Gap between an upper span line and its label.
`boxd`	3ex	Depth of lower open boxes.
`boxsep`	1.0ex	Gap between a lower open box and its label.
`labelsep`	unset	General label separation override. If set, it overrides `boxsep` and `labelsepboxes`.
`labelsepboxes`	unset	Label separation override for lower boxes.
`nestedgap`	0.3ex	Extra gap used when nested lower annotations require additional depth.
`labheight`	1.1ex	Estimated label height used in vertical reservation.
`tagy`	1.8ex	Vertical distance for floating lower tags.
`tagx`	0pt	Horizontal displacement for floating lower tags.
`underw`	0.8pt	Underline stroke width.
`autogap`	0.6ex	Extra gap used by automatic depth calculation for nested lower boxes.
`leftend`	true	Draw the left vertical end of a span or open box.
`rightend`	true	Draw the right vertical end of a span or open box.
`autodepth`	true	Let nested lower annotations increase the depth of an outer open box.
`leftarrow`	false	Add a left arrow tip to the horizontal line.
`rightarrow`	false	Add a right arrow tip to the horizontal line.
`arrowlen`	1.3ex	Arrowhead length.
`arrowtip`	Latex	TikZ arrowhead type.
`arrowscale`	1	Arrowhead scale.
`arrowinset`	0pt	Arrowhead inset.
`labelalign`	center	Label alignment for lower boxes: `left`, `center`, or `right`.

The compact reservation layer currently reserves a fixed amount of vertical space: `4.5ex` below the line and `7.5ex` above the line. This is a pragmatic setting that prevents frequent collisions in examples embedded in `gb4e`.

Example. Changing classic spacing globally inside a document

```
\setsintaxisschema{docencia}
```

Example. Changing classic spacing locally for one annotation

```
\spanline[liney=3ex,linegap=0.8ex,labsep=1.2ex,linew=1pt]{pvs}{compró manzanas}
\openbox[boxd=3.6ex,boxsep=1.4ex,linew=1pt]{od}{manzanas}
\tagbelow[tagy=2.2ex,tagx=0.5ex]{md}{las}
```

PVS

(1) compró manzanas

(2) manzanas

OD

(3) las
MD

5 Visual presets

The package option selects a visual preset for classic mode:

```
\usepackage[docencia]{sintaxismark}
\usepackage[impresion]{sintaxismark}
\usepackage[diapositivas]{sintaxismark}
\usepackage[publicacion]{sintaxismark}
```

Presets may also be selected inside the document:

```
\setsintaxisschema{diapositivas}
```

Preset	Settings
`docencia`	`linew=0.9pt, liney=2.4ex, linegap=0.6ex, labsep=1.0ex, boxd=3.2ex, boxsep=1.0ex, tagy=1.8ex, underw=1.0pt`.
`impresion`	`linew=0.5pt, liney=1.8ex, linegap=0pt, labsep=0.6ex, boxd=2.6ex, boxsep=0.8ex, tagy=1.6ex, underw=0.7pt`.
`diapositivas`	`linew=1.2pt, liney=3.0ex, linegap=0.8ex, labsep=1.2ex, boxd=3.6ex, boxsep=1.4ex, tagy=2.0ex, underw=1.2pt, arrowlen=1.8ex, arrowscale=1.25, labelsepboxes=1.6ex, nestedgap=0.8ex, labheight=1.2ex`.
`publicacion`	`linew=0.48pt, liney=2.0ex, linegap=0.3ex, labsep=0.7ex, boxd=2.8ex, boxsep=0.85ex, tagy=1.65ex, underw=0.65pt, arrowlen=1.2ex, arrowscale=1.0, labelsepboxes=0.9ex, nestedgap=0.45ex, labheight=1.0ex`.

6 Core classic commands

6.1 Upper spans: `\spanline`

```
\spanline[<keys>]{<label>}{<text>}
```

`\spanline` draws an upper horizontal line with optional vertical ends and a label above it.

Example. Upper spans

```
\spanline{ses}{Juan} \spanline{pvs}{compró manzanas}
\spanline[color=blue,linew=1pt,liney=3ex,labsep=1.2ex]{pvs}{compró manzanas}
\spanline[leftend=false,rightarrow,color=red]{pvs}{compró manzanas}
```

SES PVS
(4) Juan compró manzanas

PVS
(5) compró manzanas

(6) compró manzanas PVS

6.2 Lower open boxes: `\openbox`

```
\openbox[<keys>]{<label>}{<text>}
```

`\openbox` draws a lower open bracket-like box with a label below it.

Example. Lower open boxes

```
\openbox{od}{manzanas}
\openbox[boxd=3.8ex,boxsep=1.4ex,linew=1pt]{od}{manzanas rojas}
\openbox[labelalign=left,color=purple]{mi}{de Juan}
\openbox[rightend=false,rightarrow,color=orange!80!black]{od}{manzanas}
```

(7) manzanas
OD

(8) manzanas rojas
OD

(9) de Juan
MI

(10) manzanas
OD

6.3 Floating labels: `\tagabove` and `\tagbelow`

```
\tagabove[<keys>]{<label>}{<text>}
\tagbelow[<keys>]{<label>}{<text>}
```

Example. Floating labels

```
\tagabove{or}{Juan compró manzanas}
\tagbelow{md}{las}
\tagbelow[tagx=0.5ex,tagy=2.2ex,color=blue]{md}{las}
```

(11) Juan compró manzanas OR

(12) las MD

(13) las
MD

6.4 Underline: `\synul`

```
\synul[<keys>]{<text>}
```

Example. Underline

```
\synul{compró}
\synul[color=red,underw=1.2pt]{compró}
```

(14) compró

(15) compró

7 Shortcut commands

Upper-span shortcuts are wrappers around `\spanline`: `\SES`, `\ST`, `\SEC`, `\PVS`, `\PVC`, and `\OS`.

Example. Upper shortcut commands

```
\SES{Juan} \PVS{compró manzanas}
\OS{\SES{Juan} \PVS{compró manzanas}}
```

(16) SES PVS
Juan compró manzanas

(17) SES OS PVS
Juan compró manzanas

Many lower shortcuts are dual commands. Without a star they produce a lower tag; with a star they produce a lower open box.

```
\OD{manzanas} % lower tag + underline
\OD*{manzanas} % lower open box
\MI{de Juan} % lower tag
\MI*{de Juan} % lower open box
```

(18) manzanas
OD

(19) manzanas
OD

(20) de Juan
MI

(21) de Juan
MI

Command	Output
SES, ST, SEC, PVS, PVC, OS	Upper spans.
OD, OI	Without star: lower label with underline. With star: lower open box.
CCL, COMP, MD, MI, CAg, CREG, CSim, CL, CM	Without star: lower tag. With star: lower open box.
PSO, PSNO, POO, PONO	Without star: lower tag. With star: lower open box.

ACT, ACL, ACM, ACF, ACC, ACCant, ACI, ACFin	Without star: lower tag. With star: lower open box.
MG, Ap, MF, IPCR, Incl	Without star: lower tag. With star: lower open box.
N, NN, NV, NP, NA, NAD	Simple lower labels; NN and NV include underline.
MGV, ApV, MFV, IPCRV, InclV	Lower labels with underline.

Example. Complex shortcut example

```
\SES{\MD{El} \NN{estudiante} \MI*{de \MD{la} \N{facultad}}}
\PVS{\NV{analizó} \OD*{\MD{una} \NN{oración} \MI*{con \MD{varios} \N{modificadores}}}}
```

SES PVS

(22) El estudiante de la facultad analizó una oración con varios modificadores

MD NN MD N NV MD NN MD N

MI MI

OD

8 Manual multiline helpers

If a long span or open box must be split across visual lines, use the manual helpers. They draw separate local segments and avoid diagonal lines.

```
\spanfirst[<keys>]{<label>}{<text>} % left end, open to the right
\spanmid [<keys>]{<label>}{<text>} % continuation, no ends
\spanlast [<keys>]{<label>}{<text>} % open from the left, right end

\boxfirst[<keys>]{<label>}{<text>}
\boxmid [<keys>]{<label>}{<text>}
\boxlast [<keys>]{<label>}{<text>}
```

Example. Manual upper split

```
\spanfirst{pvs}{propuso una consigna extensa}
\spanmid{pvs}{para analizar varios casos}
\spanlast{pvs}{en la clase práctica}
```

PVS

(23) propuso una consigna extensa

PVS

para analizar varios casos

PVS
en la clase práctica

Example. Manual lower split

```
\boxfirst{od}{una consigna extensa}
\boxmid{od}{con varios incisos}
\boxlast{od}{para resolver en clase}
```

(24) una consigna extensa
OD
con varios incisos
OD
para resolver en clase
OD

The upper shortcut wrappers are also available: `\SESfirst`, `\SESmid`, `\SESlast`, `\STfirst`, `\STmid`, `\STlast`, `\SECfirst`, `\SECmid`, `\SEClast`, `\PVSfirst`, `\PVSmid`, `\PVSlast`, `\PVCfirst`, `\PVCmid`, `\PVClast`, `\OSfirst`, `\OSmid`, and `\OSlast`.

9 Line wrapping and no-break variants

`\SMlinewrap` places its contents in a variable-width paragraph of maximum width `\linewidth`. It allows long examples to wrap naturally.

```
\SMlinewrap{El \MI{departamento de unos monoblocks deteriorados} quedaba abajo.}
```

`\openboxNB` and `\spanlineNB` wrap the annotated text in `\mbox`, preventing the marked text from breaking across lines.

```
\openboxNB{mi}{departamento de unos monoblocks deteriorados}
\spanlineNB{ses}{El departamento de unos monoblocks deteriorados}
```

(25) El departamento de unos monoblocks deteriorados quedaba abajo.
MI

10 The `\breakspan` helper

`\breakspan` applies a right-opening style to all spans and boxes in its first argument and a left-opening style to all spans and boxes in its second argument.

```
\breakspan
  {\SES{\MI{El departamento de unos monoblocks}}}
  {\SES{\MI{quedaba en la planta baja}}}
```

SES SES
(26) El departamento de unos monoblocks quedaba en la planta baja
MI MI

11 Experimental automatic splitting

`\autospanline` and `\autoopenbox` require LuaLaTeX. They take plain text, compute a balanced line split, and internally emit the manual multiline helpers.

```
\autospanline[<width>]{<label>}{<plain text>}
\autoopenbox[<width>]{<label>}{<plain text>}
```

Example. Automatic split of an upper span

```
\autospanline[7cm]{pvs}{propuso una consigna extensa para analizar varios casos en la
    clase práctica}
```

PVS
propuso una consigna

PVS
extensa para analizar

PVS
varios casos en la clase práctica

Example. Automatic split of a lower box

```
\autoopenbox[7cm]{od}{una consigna extensa con varios incisos para resolver en la clase
    práctica}
```

una consigna extensa con varios
OD

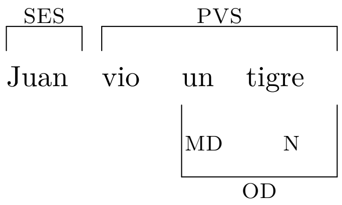
incisos para resolver en la clase práctica
OD

These commands are experimental and do not parse nested sintaxismark markup.

12 Experimental grid mode

The `sintaxisgrid` environment reads a full nested structure, extracts tokens, computes line breaks, and draws the whole analysis as a single TikZ picture. It is the preferred mode for long examples in which classic spans or boxes would cross line breaks.

```
\begin{sintaxisgrid}[width=.92\linewidth,valign=base,yoffset=0pt]
\SA{
  \SES{Juan}
  \PVS{
    vio
    \OD*{
      \MD{un}
      \N{tigre}
    }
  }
}
\end{sintaxisgrid}
```

(27) 

12.1 Grid environment options

Option	Meaning
`width=<dimension>`	Available width for line breaking. Default: `\linewidth`.
`valign=base`	Baseline alignment; recommended inside `gb4e`.
`valign=t`	Top alignment.
`valign=c`	Center alignment.
`valign=b`	Bottom alignment.
`yoffset=<dimension>`	Manual vertical adjustment. Default: `0pt`.

12.2 Grid interpretation of shortcut commands

Inside `sintaxisgrid`, commands are not executed in the classic sense. They are parsed structurally.

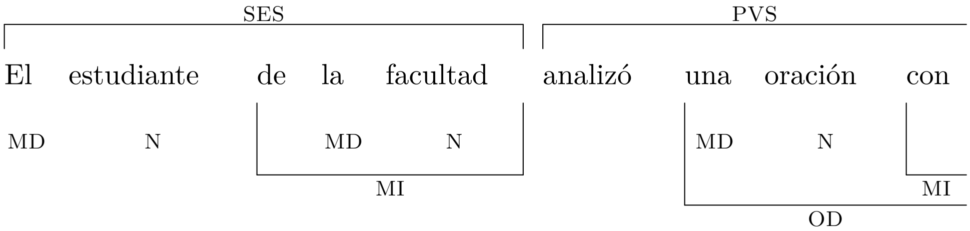
Command pattern	Grid interpretation
`SES, ST, SEC, PVS, PVC, OS`	Upper spans.
`OD, OI`	Lower underlined labels.
`Any starred shortcut, e.g. OD*, MI*, COMP*`	Lower open box.
`Other unstarred shortcuts, e.g. MD, MI, N, NN, NV, ACL`	Lower simple tags.
`SA`	Structural wrapper; it does not draw its own mark.

12.3 Grid spacing

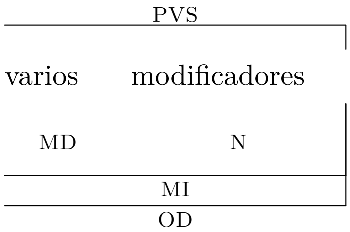
The grid renderer currently uses fixed internal constants tuned to resemble classic `\spanline` and `\openbox`: upper spans approximate `liney`/`labsep`; lower boxes approximate `boxd`/`boxsep`; lower tags approximate `tagy`; nested lower levels approximate `nestedgap` and the classic automatic depth behavior.

Example. Nested lower marks in grid mode

```
\begin{sintaxisgrid}[width=.92\linewidth,valign=base]
\SA{
  \SES{\MD{El}\N{estudiante}\MI*{de \MD{la}\N{facultad}}}
  \PVS{analizó \OD*{\MD{una}\N{oración}\MI*{con \MD{varios}\N{modificadores}}}}
}
\end{sintaxisgrid}
```

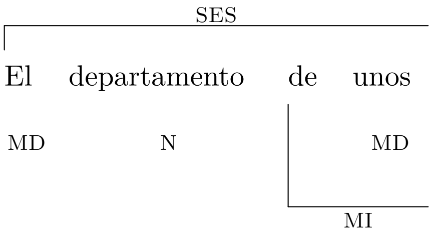


(28)

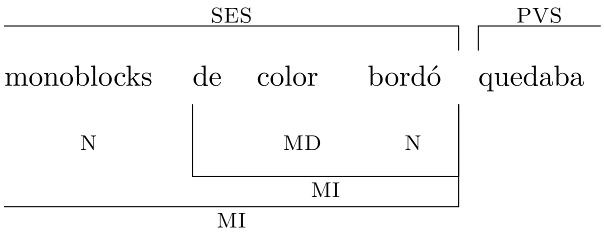


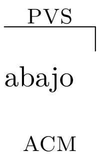
Example. Forcing a line break by narrowing the grid

```
\begin{sintaxisgrid}[width=8cm,valign=base]
\SA{
  \SES{\MD{El}\N{departamento}\MI*{de \MD{unos}\N{monoblocks}\MI*{de \MD{color}\N{bordó
      }}}}
  \PVS{quedaba \ACM{abajo}}
}
\end{sintaxisgrid}
```



(29)





Warnings such as `Missing character: There is no U+000A in font nullfont` may appear in this experimental grid version. They are not fatal and do not prevent PDF generation.

13 Recommended workflow

1. Use classic mode for short examples and for precise manual control.
2. Use `\openboxNB`, `\spanlineNB`, or the manual split helpers when an annotated classical span must not cross a line break.
3. Use `\autospanline` or `\autoopenbox` only for plain text.
4. Use `sintaxisgrid` for long examples with nested sintactic markup.

14 Version notes

- **0.7.2 experimental**: monolithic file. `sintaxismark.sty` contains classic mode, automatic split helpers, and grid mode with embedded Lua.
- **0.7.1 experimental**: fixes body capture in `sintaxisgrid`.
- **0.7.0 experimental**: introduces `sintaxisgrid`.
- **0.6.23**: adds `\autospanline` and `\autoopenbox`.
- **0.6.22**: adds manual multiline helpers.
- **0.6.21**: improves nested `\tagbelow` depth handling.